Abhilash Shah

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Education

B.Tech in Computer Science and Engineering **Expected 2028**
VIT Bhopal University *Bhopal, Madhya Pradesh*

Technical Skills

Languages:	Python, C++, JavaScript (ES6+), TypeScript, SQL, Bash, HTML, CSS
AI/ML:	Machine Learning, Deep Learning, Recommender Systems, Learning-to-Rank, NLP, Feature Engineering, Model Evaluation, LLMs, Prompt Engineering
ML Libraries:	PyTorch, TensorFlow, Keras, XGBoost, scikit-learn, Pandas, NumPy, MLflow, Jupyter
Backend:	FastAPI, Node.js, Express.js, Hono, RESTful APIs, WebSockets, Socket.IO, WebRTC, JWT, OAuth 2.0, RBAC
Frontend:	React.js, Next.js (App Router), Vite, Tailwind CSS, Responsive Design
Databases:	PostgreSQL, Supabase, MongoDB, MySQL, SQLite, SQLAlchemy (ORM)
Cloud & DevOps:	Docker, Docker Compose, CI/CD (GitHub Actions), Vercel, Oracle Cloud (OCI), Linux, Git, GitHub
Testing:	pytest, Vitest, Unit and Integration Testing, Postman
Core Concepts:	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Statistics, A/B Testing, Agile, SDLC

Projects

VIT Bhopal Sports Facilities — Full-Stack Booking & Access System **Live at VIT Bhopal** | GitHub
Next.js, TypeScript, Hono, Supabase (PostgreSQL, RLS), Tailwind CSS, Vitest, PGlite

- Deployed to **production** for **1,000+ weekly active users** across **3 sports facilities** (pickleball, tennis, turf), with an architecture designed to **scale** to the full **10,000-student** campus.
- Guaranteed **zero double bookings** under **concurrent requests** using **PostgreSQL unique constraints**, and enforced **RBAC** for **3 user roles** through **Row-Level Security** behind **Google OAuth/OTP**.
- Automated the **end-to-end booking lifecycle** (QR check-in, waitlist promotion, no-show tracking) and shipped an **admin analytics dashboard**, with access policies verified by **integration tests**.

CartSense — Explainable Product Recommendation Engine GitHub
Python, PyTorch, XGBoost, scikit-learn, MLflow, FastAPI, PostgreSQL, React, TypeScript, Docker, GitHub Actions

- Built a **two-stage hybrid recommender** on **814K real Amazon reviews** (25K users, 23K products): **4 retrieval models** (ALS, item-kNN, TF-IDF, popularity) generate candidates that an **XGBoost LambdaMART** ranker re-orders using **24 engineered features**.
- Improved **NDCG@10 by 39%** (0.0131 to 0.0182) over the strongest of **6 baselines**, including a **PyTorch Neural Collaborative Filtering** model, on a **leakage-free**, time-based held-out period, with every experiment tracked in **MLflow**.
- Designed an **A/B testing** framework that compares the model with a popularity control using a **two-proportion z-test** on click-through and add-to-cart rates, and solved **cold start** by personalizing for guests after one product view.
- Served precomputed, **explainable** recommendations through a versioned **FastAPI REST API** with **8 endpoint groups**, **JWT** and **API-key** authentication, a **retraining pipeline** driven by store events, and **CI/CD** on GitHub Actions.

MeriYatra — Real-Time Public Transit Tracking Platform GitHub
React.js, Vite, Tailwind CSS, Supabase Realtime, PostgreSQL, WebSockets, React-Leaflet, OSRM

- Architected a **real-time tracking platform** that connects daily commuters with **3 types of local transit**: autos, cabs and private buses.
- Streamed live vehicle locations over **WebSockets** with **sub-100 ms latency** and recalculated **ETAs** through an **OSRM routing engine** on every GPS update.
- Delivered **2 role-based dashboards** (driver and passenger) with ride scheduling, route search, authentication and a **bilingual (English/Hindi)** interface; packaged as a **Chrome Extension** and hosted on **Vercel**.

Experience

Open Source Contributor (Top 100), GirlScript Summer of Code (GSSoC) **2026**

- Ranked among the **Top 100 contributors** of GSSoC 2026 after merging **150+ pull requests** (features, bug fixes and documentation) across multiple **open-source** repositories.
- Ramped up quickly on unfamiliar codebases and addressed maintainer feedback through **Git-based code review**, iterating until each pull request met the project's quality standards.

Technical Team Member, 5 College Clubs, VIT Bhopal

- Contributed to the technical teams of **5 college clubs**, balancing **multiple concurrent projects** through strong **collaboration** and ownership.
- Managed technical operations for large-scale campus events with **1,000+ participants**, coordinating with **cross-functional teams** to ensure smooth execution.

Certifications

- **Oracle Cloud AI Foundations Associate**, Oracle
- **Prompt Engineering**, Infosys Springboard
- **Full Stack Web Development**, Apna College
- **Data Analytics Job Simulation**, Deloitte Australia
- **Introduction to Internet of Things**, NPTEL